

Ex. No. 3 — Designing a Quarter-Wave Transformer by Varying a) Frequency b) Input Impedance c) Characteristic Impedance

Test Case 1 — Varying Frequency

Design a quarter-wave transformer for an input impedance $Z = 50 \Omega$ and a load impedance $Z = 100 \Omega$.

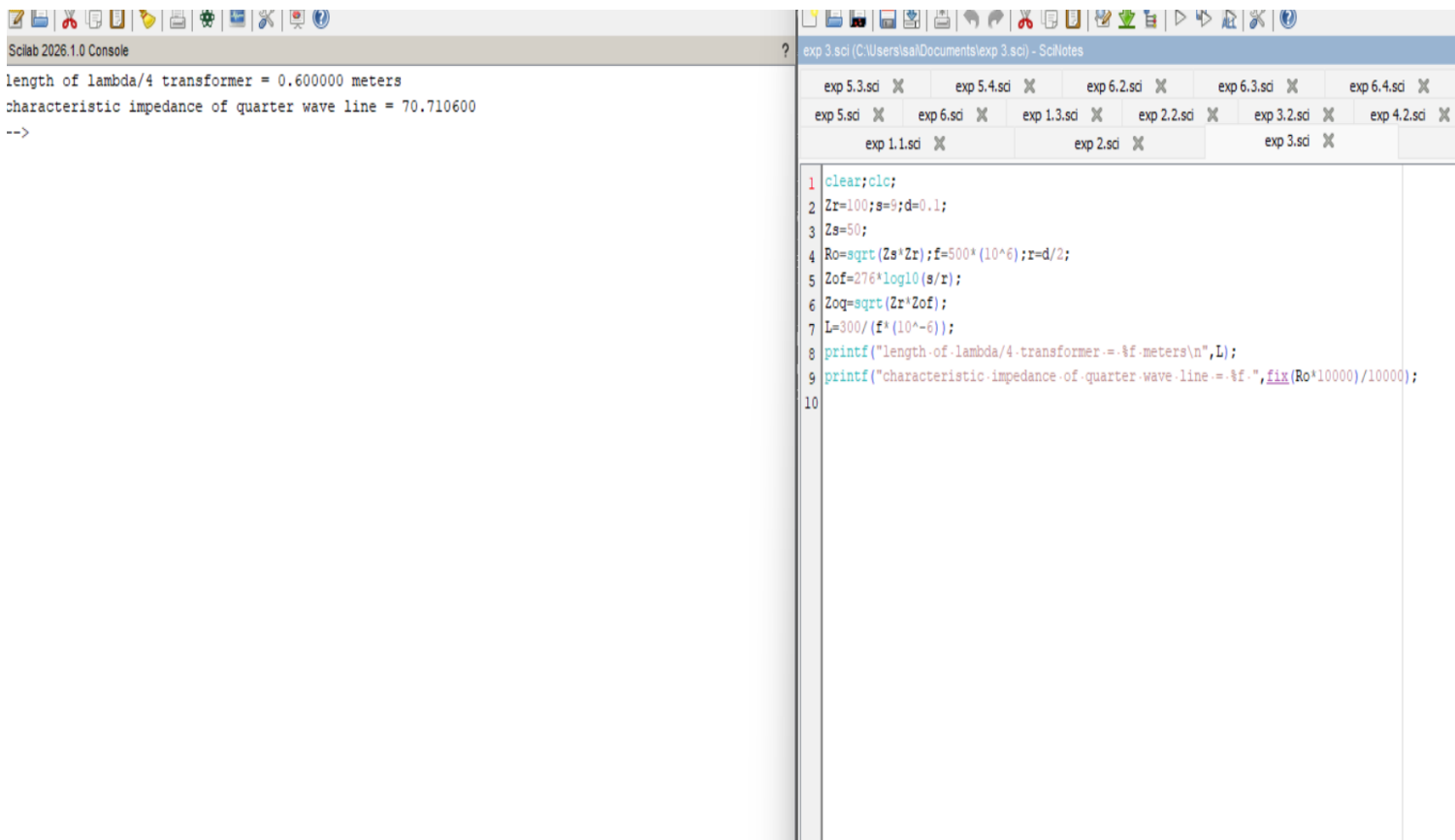
in

L

Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

Scilab Program

```
clear; clc; Zr =  
100; Zs = 50;  
f = 100*10^6; // set frequency per row (Hz) Ro  
= sqrt(Zs*Zr); L = 300/(f*10^-6); printf('length of  
lambda/4 transformer = %f meters\n', L);  
printf('characteristic impedance of quarter wave line = %f ', fix(Ro*10000)/10000);
```



Sample Console Output

```
length of lambda/4 transformer = 0.600000 meters
characteristic impedance of quarter wave line = 70.710600
-->
```

Characteristic Impedance of the Quarter-Wave Line

Z _L (ohms)	Z _{in} (ohms)	Simulation R _{in} (ohms)	Theoretical R _{in} (ohms)
50	100	70.7107	70.7107

Length of λ/4 Transformer vs Frequency (Z = 100 Ω, Z = 50 Ω) L_{in}

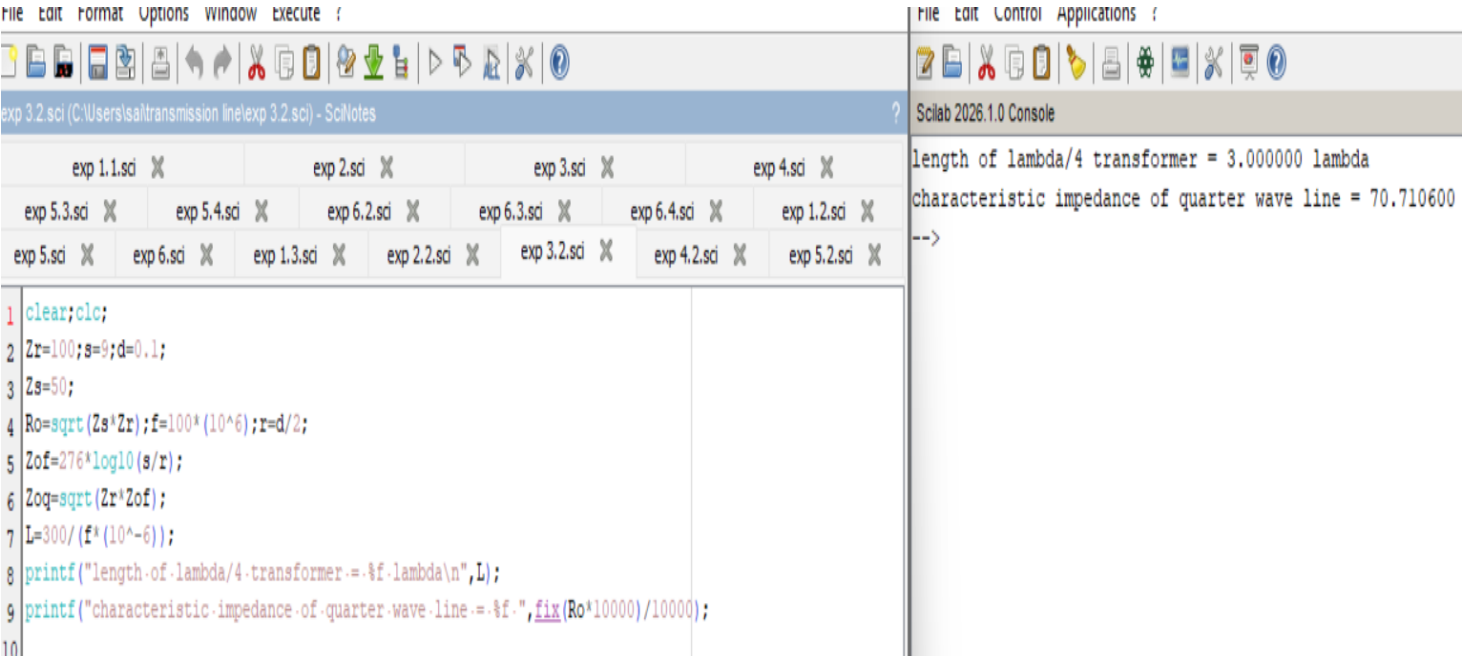
Frequency (MHz)	Simulation Length (m)	Theoretical Length (m)
100	3.0000	3.0000
200	1.5000	1.5000
300	1.0000	1.0000
500	0.6000	0.6000

Test Case 2 — Varying Input Impedance

For a fixed operating frequency of 2 GHz and a load impedance $Z = 100\ \Omega$, design a quarter-wave transformer L and evaluate its performance as the input impedance Z varies through $\{25\ \Omega, 50\ \Omega, 75\ \Omega, 100\ \Omega\}$. Validate in that the transformer's characteristic impedance ($Z_o = R_o = \sqrt{(Z \cdot Z_L)}$) changes with Z . transformer in

Scilab Program

```
clear;clc; Zr = 100;
f = 2*10^9;
Zs = 25; // set Zin per row
Ro = sqrt(Zs*Zr); L = 300/(f*10^-6); printf('length of
lambda/4 transformer = %f lambda\n', L);
printf('characteristic impedance of quarter wave line = %f ', fix(Ro*10000)/10000);
```



Sample Console Output

```
length of lambda/4 transformer = 3.000000 lambda
characteristic impedance of quarter wave line = 70.710600
-->
```

R_{in} vs Z_L (Z = 100 Ω, f = 2 GHz) in

Z _L (ohms)	Simulation R _{in} (ohms)	Theoretical R _{in} (ohms)
25	50.0000	50.0000
50	70.7107	70.7107
75	86.6025	86.6025
100	100.0000	100.0000

Result A quarter-wave transformer is designed at a fixed operating frequency of 2 GHz and load impedance (Z_L) of 100 Ω, and its performance is evaluated for various input impedance values. The variation of the transformer's characteristic impedance with Z_L is also validated.